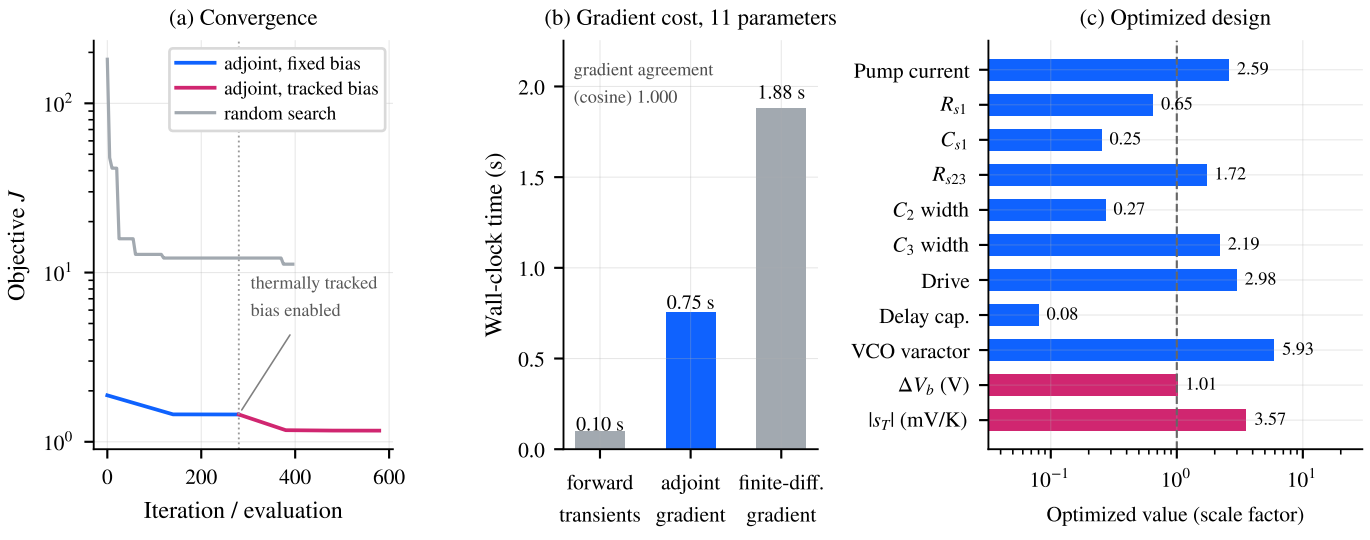
(a) Convergence
adjoint, fixed bias
adjoint, tracked bias
random search
thermally tracked
bias enabled
Objective J
Iteration / evaluation
(b) Gradient cost, 11 parameters
gradient agreement
(cosine) 1.000
Wall-clock time (s)
0.10 s
0.75 s
1.88 s
forward
transients
adjoint
gradient
finite-diff.
gradient
(c) Optimized design
Pump current
R_{s1}
C_{s1}
R_{s23}
C_2 width
C_3 width
Drive
Delay cap.
VCO varactor
ΔV_b (V)
$|s_T|$ (mV/K)
2.59
0.65
0.25
1.72
0.27
2.19
2.98
0.08
5.93
1.01
3.57
Optimized value (scale factor)